

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 9: Local Area Networks (LANs) & Ethernet (CSMA/CD)

Course: Computer Networks (Credit Hours: 3)

Instructor: Muhammad Usman Ghani

1. WHAT IS A LOCAL AREA NETWORK (LAN)?

A **Local Area Network (LAN)** is a computer network confined to a limited geographical area, such as a single classroom, computer lab, office building, or university campus.

Key Characteristics: High data transfer speeds (100 Mbps to 10 Gbps), low setup costs, low propagation delay, and ownership by a single organization.

2. WHY DO WE NEED MAC (MEDIUM ACCESS CONTROL)?

In a traditional shared LAN, multiple computers are connected to the exact same transmission wire or channel. If two computers transmit data at the exact same moment, their signals collide and corrupt each other (a **Collision**). MAC protocols provide rules to prevent or manage collisions.

Easy Everyday Example A classroom conversation: If everyone starts shouting at once, nobody understands anything. The rule is: *listen first*, wait until the other person stops speaking, and then talk!

3. TRADITIONAL ETHERNET & CSMA/CD

Ethernet (IEEE 802.3) is the most widely installed LAN technology in the world. Classic shared Ethernet uses **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection).

Understanding the Name

- **Carrier Sense (CS):** Listen before talking. A station checks if the cable is busy or idle before sending data.
- **Multiple Access (MA):** Many computers share the exact same wire.
- **Collision Detection (CD):** While sending, the station continues listening. If it detects a voltage spike, it knows a collision occurred!

What Happens During a Collision?

- The transmitting station immediately stops sending.
- It broadcasts a **Jam Signal** to alert all other nodes to stop transmitting.
- The station waits for a random period (using the **Exponential Backoff Algorithm**) and tries to retransmit.

4. ETHERNET FRAME FORMAT & MAC ADDRESSING

Field Name	Size	Purpose
Preamble & SFD	8 Bytes	Alternating pattern of 1s and 0s to synchronize receiver clock timing.
Destination MAC	6 Bytes (48 bits)	Physical hardware address of the receiving device (e.g., 00:1A:2B:3C:4D:5E).
Source MAC	6 Bytes (48 bits)	Physical hardware address of the transmitting device.
Type / Length	2 Bytes	Specifies the upper-layer protocol (e.g., IPv4 or ARP).
Data (Payload)	46 to 1500 Bytes	Actual packet delivered from the Network Layer.
FCS (CRC)	4 Bytes	Cyclic Redundancy Check to detect corrupted bits.

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why is CSMA/CD not needed in modern Gigabit Full-Duplex Ethernet?

Ans: Because modern networks use dedicated micro-segmented switch ports with separate transmit (TX) and receive (RX) wire pairs, completely eliminating collisions.

Q2: What is the size of a MAC address in bits and bytes?

Ans: **48 bits** or **6 bytes**, typically written as 12 hexadecimal digits.